



Why do some countries receive more international financing for coal-fired power plants than renewables? Influencing factors in 23 countries

Achmed Edianto ^{a,*}, Gregory Trencher ^b, Kazuyo Matsubae ^a

^a Graduate School of Environmental Studies, Tohoku University, Sendai 980-8577, Japan

^b Graduate School of Global Environmental Studies, Kyoto University, Kyoto 606-8501, Japan

ARTICLE INFO

Article history:

Received 13 August 2021

Revised 15 October 2021

Accepted 9 December 2021

Available online 29 December 2021

Keywords:

Coal power

Electricity

International investments

Public finance

Renewables

Coincidence analysis

ABSTRACT

Construction of new coal-fired power plants (CFPPs) continues around the world, jeopardizing efforts to mitigate climate change. In 2020, some 500 GW of projects are under construction or being planned, mostly in developing countries. Because of high costs and technological hurdles, construction in developing countries is largely assisted by public finance from richer nations. Troublingly, in recent years, financing has provided more support to coal-fired electricity than to renewables. An important question thus arises: What factors drive the flow of international financing toward coal or renewables? Prior research has not tackled this question systematically with quantitative methods and a large sample of countries. This study thus examines 23 countries receiving international financing from government-affiliated bilateral institutions for both coal and renewables. We use a novel “coincidence analysis” method to systematically identify the factors that explain why some countries received more investment for CFPPs than renewables. Two combinations of influencing factors are identified: countries with: (i) a bilateral agreement to promote coal industry development, and weak or absent carbon pricing; or (ii) weak policy support for renewable energy, and weak or absent carbon pricing. We explore the influence of these factors in four countries—Indonesia, Pakistan, Mongolia, and Vietnam—to deepen understanding of why some countries receive higher investment for coal than renewables. This analysis leads to important policy implications.

© 2021 The Author(s). Published by Elsevier Inc. on behalf of International Energy Initiative. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Introduction

Coal combustion is the single biggest contributor to anthropogenic CO₂ emissions and global heating (Edenhofer et al., 2018; IPCC, 2018; Oberschelp et al., 2019). To mitigate dangerous climate change in line with the Paris Agreement's objectives, there is a need to both reduce coal combustion and accelerate the deployment of renewable energy (Nace, 2018; Tollefson, 2018). Despite a recent slowdown, in 2020 some 500 GW of coal-fired power plants (CFPPs) were under construction or being planned around the world (Global Energy Monitor, 2021a). In addition to China and India, most new construction is in developing countries such as Indonesia, Vietnam, Bangladesh, South Africa, the Philippines, and Mongolia. Although they are blessed with attractive geographical conditions, diverse obstacles have hampered the shift to renewables-based power generation in developing countries. These obstacles include a lack of supportive policies and institutions at national and regional level, and inferior technological, economic, and

business capabilities relative to wealthier or more experienced countries (Sen & Ganguly, 2017; Shukla et al., 2017). In such conditions, many developing nations have opted to develop their economy and expand their power-generation capacity through coal (Steckel et al., 2015).

Because of the need for massive capital outlay (often reaching several billion dollars) and technological complexity (Lin & Bega, 2021), construction of CFPPs in developing countries is often assisted by developed countries. Financial support is typically received through public (i.e. government-affiliated) institutions such as development banks and export credit agencies (Chen et al., 2016; Uddin et al., 2019) as well as private-sector banks and companies. Developing countries also receive technological support from electric utilities and plant-equipment manufacturers in richer countries, which supply engineering and design expertise, often taking charge of construction, operation, and maintenance (Trencher, Downie et al., 2020). Of the many countries carrying out such roles, financing from China, Japan, and South Korea has attracted much attention (Han & Jack, 2017; Mori, 2018; Trencher, Downie et al., 2020; Zhao & Alexandroff, 2019).

The urgent need to reduce CO₂ emissions and air pollution in developing economies (Kopplitz et al., 2017) thus requires that investments and

* Corresponding author.

E-mail address: achmed.shahram.edianto.p1@dc.tohoku.ac.jp (A. Edianto).

transfers of power-sector technology and expertise from wealthy nations flow toward renewables and other clean energy sources (Manych et al., 2021). Yet in recent years, international public financing (Han & Jack, 2017; Uddin et al., 2019; Zhao & Alexandroff, 2019)—which makes up the majority of financing in the power sector in developing countries—has supported coal rather than renewables. Indeed, in 2013–2017, the latest period for which data exists, overseas financing for coal from government-affiliated financial institutions totaled \$55 billion,¹ more than three times the \$18 billion provided for renewable energy (Han & Jack, 2017). Over 90% of this financing for coal came from just three countries: China (55%), Japan (29%), and South Korea (9%). Although international financing for coal has since decreased (IEEFA, 2021), CFPPs typically operate over multiple decades. Their operational lifetime averages 40–50 years globally (Erickson et al., 2015), and many are protected by multi-decade purchase agreements providing stable income to investors (Trencher, Downie et al., 2020). Thus, decarbonization efforts in countries building new CFPPs will be negatively impacted by the CO₂ emissions that are locked in over the lifetime of each asset's operation (Jewell et al., 2019; Steckel et al., 2015).

This situation prompts a need to study the factors affecting the flow of financing toward coal or renewables, especially in countries receiving both. Many relevant factors are discussed in literature, but this knowledge remains scattered across different studies and academic fields. For example, high production costs (Bridle et al., 2018; Ialnazov & Keeley, 2020) and a lack of supportive government institutions (Gabriel, 2016) have been shown to deter investments in renewable energy. Meanwhile, conditions that may encourage CFPP development or prolong reliance on coal power include the high rates of economic growth and rising energy consumption typically observed in low income countries (Kurniawan et al., 2020; Steckel et al., 2015), the availability of local coal reserves (Keeley & Matsumoto, 2018a; Blondeel et al., 2020; Diluiso et al., 2021) as well as weak environmental governance and political transparency (Blondeel et al., 2020; Jewell et al., 2019; Rentier et al., 2019). Moreover, few if any studies have attempted to systematically identify statistical correlations between the factors in recipient countries that affect the volume of financial flow toward both coal and renewables. So it remains unclear why some countries receive more financing for coal, while others receive more for renewables.

To address this gap, this study aims to elucidate the factors that encourage financing in coal and renewables in countries receiving international finance for power-generation projects. To pursue this objective, first we develop a framework to formulate causal expectations derived from a review of existing literature. Then we identify the most relevant influencing factors via an expert survey. We apply the resulting framework to a sample of 23 countries receiving international financing for both coal and renewables. We use a novel method called “coincidence analysis” to mathematically identify the combination of factors that best explain why some countries received more international financing for coal than renewables. Finally, we focus on a sub-sample of four countries—Indonesia, Pakistan, Mongolia, and Vietnam—to qualitatively explore the influence of the identified factors with secondary documents.

This study makes several contributions to existing literature. First, we focus on the perspective of recipient countries. This fills a gap because existing research has focused on investor countries (Chen et al., 2020a; Gallagher et al., 2018; Gallagher & Qi, 2018; Koppenborg, 2016). Knowledge of recipient-side factors that drive CFPP financing has thus been underdeveloped. Second, although research exists on overseas finance and investments in the power sector from government-affiliated institutions (Chen et al., 2020a; Gallagher, 2018; Gallagher et al., 2018), this has focused on China. Here we expand the scope of research to investigate all international financing from any government-affiliated institution with a sample of 23 recipient countries. We identify the factors that

have helped other developing countries to avoid increasing reliance on coal power, generating important cues for global decarbonization efforts. Third, our use of the coincidence analysis method (Janzwood, 2020; Schmid & Bornemann, 2019) has not featured before in energy studies. This method allows us to mathematically identify correlations between a larger number of factors affecting power-sector financing than more-established methods such as qualitative comparative analysis.

Methods

Scope and overall study design

This study focuses explicitly on public financing for coal and renewables in the co-incidence analysis, but examines broader investment trends when constructing our analytical framework and the qualitative analysis on Indonesia, Pakistan, Mongolia, and Vietnam. In defining “financing”, we follow Chen et al. (2020b) and Manych et al. (2021) to include loans, guarantees (i.e. insurance) and other underwritings (e.g. the issuing of shares and bonds) provided by financial institutions, such as banks and export credit agencies. Meanwhile, we use “investment” as a larger umbrella term to refer to both public and private financing and other acts of profit generation. This includes the acquisition of equity via shares and bonds as well as the construction and operation of CFPPs (Steffen & Schmidt, 2019).

The study design consists of five steps. First, we constructed an analytical framework to identify factors from literature that positively or negatively influence overseas power-sector financing and investment for coal and renewables. Second, we gathered experts' opinions through a survey to identify the most important influencing factors, eliminating the least important ones. This step was necessitated by the coincidence analysis method, which requires an optimal balance between the number of cases and the number of influencing factors examined. Third, we built a sample of countries that received international financing for both coal and renewables from government-affiliated institutions.² Fourth, we used coincidence analysis to determine the factors that best explain why certain recipient countries in the sample received more financing for coal than renewables. Last, we enriched the analysis by sourcing qualitative evidence from secondary documents to show the influence of the identified factors in a smaller subset of countries. The first four steps are elaborated below.

Development of preliminary framework

To identify relevant studies for the analytical framework that categorizes factors that influence international financing for coal and renewables, we used Scopus and Google Scholar to search for academic and grey literature with varying combinations of keywords. These included “coal,” “renewables,” “power,” “development,” “overseas,” “international,” “driver,” “influence,” “investment,” and “financing.” Due to the limited volume of literature dealing exclusively with foreign financing or investments for CFPP, we expanded the focus to also consider factors that influence the development of renewables when we assume this to have an influence on the utilization of coal.

Analysis of this literature identified 20 factors that may influence international financing and investing for coal and renewables. This preliminary framework appears in Table 1. Building on the work of Keeley and Matsumoto (2018a), we organized these into four categories: (i) policy, (ii) political and societal, (iii) energy resources, and (iv) economic.

Expert survey

The expert survey targeted researchers and practitioners working in research institutions, NGOs, private firms, and government-affiliated

¹ All dollar units in this paper refer to US currency.

² This refers to a company or institution fully or partially owned by the government, such as development banks, commercial banks, export credit agencies, etc.

Table 1
Preliminary framework of factors influencing coal power investments as identified from the literature review.

	Influencing factors	Assumed influence	Literature
1. Policy	1.1 Renewable energy policy	Policies that support renewable energy would drive market growth and help lower costs, thus decreasing the economic advantage of CFPPs.	(Keeley & Matsumoto, 2018a,b; Ragosa & Warren, 2019)
	1.2 Incentives for international independent power producers	Incentive policies that protect baseload power sources (e.g. capacity payments, take or pay clauses) will reduce financial risk for the investor and improve prospects for profitability, thus encouraging investments in large baseloads like CFPPs.	(Trencher, Downie et al., 2020; Yildizhan, 2017)
	1.3 Emission standards	Strict emission standards increase the cost of plant construction or operation due to the need for expensive anti-pollution control equipment, thus discouraging investments in CFPPs.	(Koplitiz et al., 2017)
	1.4 GHG emissions reduction target	Weak targets to reduce GHG emissions can encourage the construction of carbon intensive infrastructure, including CFPPs, since there is no policy imperative to discourage such investments and promote alternatives.	(Bertram et al., 2015; Guzel & Okumus, 2020; Shen et al., 2019; Skoczkowski et al., 2018)
	1.5 Bilateral agreements	Bilateral agreements that emphasize coal industry development will encourage investments in coal-related infrastructure as well as the formation of supportive policies and institutional arrangements in both recipient and investor countries.	(Kox & Rojas-Romagosa, 2020; Tritto, 2021; Umbach & Yu, 2010; Yao et al., 2018; Yuan et al., 2018)
	1.6 Carbon price adoption	Carbon price policies increase the production cost of fossil fuel energy generation, reducing the competitiveness and attractiveness of investments in carbon-intensive CFPPs.	(Böhringer et al., 2017; Cole, 2004; Lamb & Minx, 2020; Mori, 2018; Wall et al., 2019; Zhang et al., 2019)
	1.7 Energy mix target	A long-term policy commitment to using coal as an energy or electricity source sends positive signals to investors, thus encouraging investments in CFPPs.	(Bertram et al., 2015; Trencher, Rinscheid et al., 2020)
2. Political and societal	2.1 Environmental awareness	Low public awareness or engagement with environmental issues would manifest as low demand for carbon-free electricity and the implementation of policies that restrict carbon-intensive energy sources or encourage alternatives.	(Chapman & Itaoka, 2018)
	2.2 Democratic functioning of government	Low democratic conditions, the presence of corruption, and low transparency or openness in decision-making can allow politicians and policymakers to form close ties with powerful fossil fuel companies with vested interests.	(Alova et al., 2021; Ialnazov & Keeley, 2020; Jewell et al., 2019; Keeley & Matsumoto, 2018a; Rentier et al., 2019)
	2.3 Political or socio-economic stability	Unstable political or socio-economic conditions create high uncertainty for investors, especially for technologies like renewable energy with long-pay back periods and the need for supportive policies. In such circumstances, CFPPs typically involve shorter payback periods and more institutional support (see 1.2).	(Dorsey, 2019; Keeley & Matsumoto, 2018a; Ragosa & Warren, 2019)
3. Energy resources	3.1 Coal resources	The availability of recoverable coal reserves in a country increases the attractiveness of coal as an electricity source for energy security and cost reasons.	(Blondeel et al., 2020; Bohlmann et al., 2019; Edenhofer et al., 2018; Keeley & Matsumoto, 2018a; Rentier et al., 2019; Steckel et al., 2015)
	3.2 Access to natural gas	The absence of local gas reserves or import facilities will prevent countries from using this energy to support renewable energy (as a back-up) or as an alternative to coal (as a baseload).	(Mac Kinnon et al., 2018; Trencher, Rinscheid et al., 2020)
4. Economic	4.1 Exchange rate	An unstable exchange increases risks for investments with long-term payback periods such as renewables. Conversely, it increases the attractiveness of government-backed infrastructure such as CFPPs, which have comparably faster paybacks.	(Keeley & Matsumoto, 2018a)
	4.2 National income	Poor economic conditions in a country create a need for cheap energy sources and hamper the development of expensive grid infrastructure to support renewables, thus encouraging investment in CFPPs.	(Gallagher et al., 2021; Kurniawan et al., 2020; Oberschelp et al., 2019; Steckel et al., 2015)
	4.3 Access to capital	Limited access to financial capital for renewable energy will challenge the market entry of new players, creating an institutional advantage for incumbent energy producers.	(Best, 2017; Zeng et al., 2017)
	4.4 Electrification rate	Low electrification rate would drive the urgency to accelerate the electricity supply, and CFPPs will be chosen because they are well established and reliable, and can be built in large size and increase the electrification ratio faster. This situation could increase the likelihood that investors and policymakers would choose coal power over other options.	(Peabody, 2017; Yang & Urpelainen, 2019)
	4.5 Electricity growth	A high growth of electricity demand creates a need to expand grids and power production, creating an imperative for affordable and proven technologies with a large generation capacity.	(Koplitiz et al., 2017; Kurniawan et al., 2020; Rasheed et al., 2021)
	4.6 Emissions intensity	An economy with a high carbon emissions intensity indicates the presence of fossil-fuel-centric industries. When such industry is supported by established policy and institutional arrangements, a country may lack resistance toward investments in carbon-intensive technologies, such as CFPPs.	(Cole, 2004; Guzel & Okumus, 2020)
	4.7 Cost of renewables generation	A high average cost of renewable power generation tends to hamper investments, increasing the attractiveness of cheaper technologies such as CFPPs.	(Bridle et al., 2018; Ialnazov & Keeley, 2020)
	4.8 Cost of coal power generation	Low average cost of coal-fired electricity increases its economic and political attractiveness, making renewables less competitive for investors and policymakers.	(Kurniawan et al., 2020; Svobodova et al., 2020)

Table 2
Overview of expert survey respondents.

Sector	Example organizations	Country	No. of responses	Portion of total (%)
NGO	Agora Energiewende, Friends of the Earth, Greenpeace, Global Energy Monitor, Solution for Our Climate, Urgewald	Germany, Japan, South Korea, USA	7	24%
Universities, think tanks and research institutes	Mercator Research Institute on Global Commons and Climate Change (MCC), North China Electric Power University, Technische Universität Berlin, Purnomo Yusgiantoro Center (PYC)	Germany, China, Indonesia	11	38%
Private companies and private research institute	Institute of Energy Economic Japan (IEEJ), JERA, Mizuho Bank, Shizen Energy, Siemens Energy	Japan, Singapore (Asia-Pacific)	8	28%
Government-affiliated organizations	Indonesia government ministry, ^a Japanese export credit agencies ^a	Indonesia, Japan	3	10%
Total			29	100%

^a Note: Exact names for organizations not wishing to disclose their names are not listed.

institutions from different countries representing major investor and recipient countries. This included China, Germany, Japan, Indonesia, Singapore, South Korea, and the USA. All possessed specialist knowledge and experience in the energy or power sector, particularly renewables and CFPPs. Experts were identified from the author names and affiliations listed in relevant publications, from referrals by other experts, and by contacting organizations to request introductions.

The survey was administered online using Google Forms from June 11 to December 31, 2020 (see Supplementary material 1). In total, 62 experts were contacted to ask permission to send the survey; 29 of these (47%) agreed, and all completed the survey. A summary of the survey respondents is shown in Table 2. While it is likely that their country of activity (see the third column) has affected responses, the 29 experts completing the survey nevertheless represent diverse viewpoints and sectors from major financing and recipient countries.

Surveys were created by converting the 20 influencing factors from the preliminary analytical framework (Table 1) into questions and accompanying explanations. Respondents were given a neutral explanation of each factor and were asked: “Could this factor increase the chances of international investments being made for coal power rather than renewables?” Respondents were prompted to assign a score using a 5-point Likert scale.

A threshold was set to eliminate any influencing factors that were deemed less important from further consideration in the empirical analysis. Factors were kept only when more than 50% of answers consisted of “highly possible” or “partly possible.” The full survey results are in Supplementary material 2.

Final analytical framework based on results of expert survey

Shown in Table 3 below, 16 of the 20 influencing factors identified from literature were approved in the expert survey—that is, they received a combined score above 50% for “highly possible” and “partly possible.” The full results of the expert survey are listed in Supplementary material 2 (Table S2.1). Of the 16 factors approved, however, only 10 are included in the final coincidence analysis, since inclusion required the availability of relevant data for the entire study period (2013–2017) and for all sampled countries. We also summarize, in Supplementary material 2 (after Table S2.1), the reasons that experts rejected four influencing factors, which we duly omitted from our analysis.

In the following sections, for the influencing factors approved by experts, we explain in detail the data sources, and any units or data-processing decisions used to dichotomize variables and construct the crisp set truth table. Readers can find a more detailed explanation in Supplementary material 3 of the basis in literature for each of the influencing factors included in the coincidence analysis (i.e. those approved by experts and for which data is available).

Renewable-energy policy (RE)

The presence of this factor (RE) is measured with World Bank data “Regulatory Indicators for Sustainable Energy 2017” (World Bank,

2017). It uses three classifications: high, medium, and low, with “high” being the most attractive for renewable energy. The original data measures policy support based on seven indicators: legal frameworks, planning for expansion, incentives and regulatory supports, financial incentives, network connections, counterparty risks, and carbon pricing. Since the presence of carbon pricing is examined individually, scoring for this indicator is omitted for this influencing factor. In constructing the truth table, we assign a score of “1” to countries classified as “high,” and “0” to “medium” and “low” countries, based on the average score of the remaining six indicators.

GHG emissions reduction target (GHG)

Coding of this factor (GHG) is based on the strength of ambition in a country's target to reduce their emissions by 2030. We extracted relevant data from the summary of Nationally Determined Contribution (NDC)³ provided by Carbon Brief (2015) to build an original dataset. Countries with an unclear target or a baseline based on business-as-usual emissions trajectories were deemed to indicate weak ambition and were categorized as “0.” Countries with clear commitments to reducing absolute emissions were categorized as “1.”

Bilateral agreement (BA)

We measure the influence of this factor (BA) based on the presence of a relevant agreement between the recipient and investor countries. We limit our analysis to agreements concluded between the recipient country and investing countries covered in the NRDC database (NRDC, 2017) in 2010–2017. Setting this period allows us to consider the influence of agreements concluded a few years before our study period of 2013–2017. Bilateral agreements were found with online searches using the name of the recipient or investor country, “coal,” and combinations of “cooperation,” “agreement,” “partnership,” “memorandum,” and “MoU.” We consider evidence in the form of actual agreements, if publicly available, and announcements of agreements on government or third-party websites. We include bilateral agreements focused on developing the coal industry (e.g. extraction and the power sector) in addition to general pacts of economic cooperation that specifically emphasize developing the coal industry.

Countries with a bilateral agreement mentioning the coal industry were assigned a score of “1.” Countries for which no relevant bilateral agreements were found were assigned “0.”

Carbon price adoption (CP)

The presence of this factor (CP) is measured with the “carbon pricing and monitoring” indicator from the above-mentioned

³ Nationally determined contributions (NDCs) are non-legally-binding policy documents submitted to the United Nations Framework Convention on Climate Change (UNFCCC) when a country ratifies the Paris Agreement (UNFCCC, 2015). These outline the actions and policy measures that a country intends to take to achieve the self-determined GHG emissions reduction target after 2020 in addition to adaptation measures.

Table 3
Summary of final framework and data considerations.

	Influencing factors	Approval by experts	Proposed indicator	Data availability	Data source ^a	Unit in original dataset	Used in coincidence analysis
Policy	Renewable energy policy	Yes	The quality of policies supporting renewable energy.	Yes	World Bank (2017)	Low/medium/high	Yes
	Incentives for international independent power producers	Yes	The availability of incentives such as long-term power purchase agreements, take-or-pay clauses, etc.	No	–	–	No
	Emission standards	Yes	Regulatory emission limits for atmospheric pollutants (e.g. CO ₂ , SO ₂ , NO _x , etc.).	No	–	–	No
	GHG emissions reduction target	Yes	The ambition level of emission reduction targets stated in Nationally Determined Contributions (NDC).	Yes	Carbon Brief (2015)	Weak/strong	Yes
	Bilateral agreements	Yes	The presence of bilateral agreements forged with an investing country that mention developing the coal industry	Yes (built by author)	Original data	Yes/no	Yes
	Carbon price adoption	Yes	The strength of carbon pricing regulation, as measured by the World Bank's "Regulatory Indicators for Sustainable Energy".	Yes	World Bank (2017)	Low/medium/high	Yes
	Energy mix target	Yes	An ongoing commitment to utilize coal power, as stated in Nationally Determined Contributions (NDC)	Yes	UNFCCC (2020)	Yes/no	Yes
Political and societal	Environmental awareness	Yes	The level of societal awareness on environmental or climate issues, as measured by public surveys.	Not for entire sample	Pelham (2009)	Score 1–100	No
	Democratic functioning of government	Yes	Rankings of electoral process, independent media, civil society participation, judicial systems, and corruption, as measured by democracy scores in the "Freedom Level" database, published by US-based NGO Freedom of House.	Yes	Freedom of House (2018)	Free/partially free/not free	Yes
	Political or socio-economic stability	No	The stability of political and socio-economic conditions.	No	–	–	No
Energy resources	Coal resources	Yes	The volume of annual per capita coal production.	Yes	IEA (2020)	KtoE per capita/year	Yes
	Access to natural gas	Yes	The average share of gas generation in the power mix.	Yes	IEA (2020)	Share of annual electricity generated by gas (%)	Yes
Economic	Exchange rate	No	Volatility of a country's currency compared to other major currencies.	Yes	–	–	No
	National income	Yes	The classification of a country based on national income rankings, as measured by World Bank data.	Yes	World Bank (2019)	High income/upper middle/ lower middle/low income	Yes
	Access to capital	Yes	The level of accessibility to capital for renewable energy.	No	–	–	No
	Electrification rate	No	The percentage of a country's population with access to electricity.	Yes	–	–	No
	Electricity growth	Yes	The annual growth of electricity demand, as measured by World Bank data.	Yes	World Bank (2019)	Annual growth (%)	Yes
	Emission intensity	No	The volume of CO ₂ emissions produced for each unit of economic output.	Yes	–	–	No
	Cost of renewables generation	Yes	The levelized cost of producing renewable electricity per kilowatt hour.	Not for entire sample	–	–	No
	Cost of coal power generation	Yes	The levelized cost of producing coal-fired electricity kilowatt hour.	Not for entire sample	–	–	No

^a Note: Data sources only listed for influencing factors examined in the coincidence analysis.

World Bank data (World Bank, 2017). Following the *Renewable-energy policy (RE)*, in constructing the truth table, we assign "1" to countries classified as "high," and "0" to those deemed "medium" or "low."

Energy mix target (NDC)

The coding of this factor (NDC) is based on the presence of ambitions to use coal as a source of electricity in the international climate pledges contained in the first Nationally Determined Contributions (NDC) submitted to the United Nations (UNFCCC) (2020) in 2016 and 2017. Countries that mention a commitment to using coal in their future energy mix were assigned a score of "1." Those that did not mention this were assigned "0."

Democratic functioning of the government (DG)

Data for this influencing factor (DG) were sourced from the "democracy scores" in the "Freedom Level" database published by the US-based NGO Freedom of House (2018). Countries identified as "free" were assigned a score of "1," while those receiving "partially free" or "not free" were assigned "0."

Coal resources (CR)

The presence of this factor (CR) is measured by the amount of annual per capita coal production in a country, compared to the world average. Data for 2017 was extracted from the "Data and Statistics" database from the International Energy Agency (IEA, 2020). Countries with an annual per capita production volume equal to or above the world

average were assigned a score of “1.” Countries below the world average were assigned “0.”

Access to natural gas (NG)

Coding of this factor (NG) is measured by the share of gas in each country's electricity mix. This reflects the assumption that countries with a low utilization rate would have limited domestic resources or infrastructure for importing. Data for 2017 is derived from the “Data and Statistics” database from the International Energy Agency (IEA, 2020). Countries were assigned “1” if gas was the first or second principal source of electricity, or, if not, if it was above 20% of total power generation. Those not meeting either condition were assigned “0.”

National income (NI)

Coding of this factor (NI) is measured from the World Bank's (2019) country classifications for gross national income per capita. Countries identified as “high-income” were assigned a score of “1.” Those classified outside this, falling between “upper-middle income” and “low-income,” were assigned “0.”

Electricity growth (EG)

Coding of this factor (EG) is based on a country's average growth of electricity demand compared to the world average over 2008–2013. Data was obtained from the World Bank database “Electric Power Consumption” (World Bank Group, 2019). Countries with a growth above the world average were given a score of “1.” Those below this were given “0.”

Table 4 shows the final truth table for all influencing factors based on the binary scoring explained above. The raw data for all factors can be found in Supplementary material 4 (Tables S4.1–4.10).

Source of financial data and sample construction

We built our sample of countries receiving both financing for coal and renewables with data extracted from the Natural Resources Defense Council (NRDC) database (NRDC, 2017). This covers the period from January 2013 to August 2017 and contains financing amounts awarded by G20 public institutions to international coal and renewable

energy projects. Concretely, this data includes direct project financing, guarantees for projects (i.e. export insurance), and loans to financial intermediaries from institutions, such as bilateral and multilateral development banks, as well as export credit and insurance agencies. We excluded multilateral development banks from our analysis because for coal financing, the vast majority was provided by bilateral institutions (around \$64 billion in total compared to only \$1 billion from multilateral). In addition, for renewables, the amount of financing from multilateral institutions is higher than bilateral (\$24 billion compared to \$18 billion). This trend indicates that multilateral banks have recently shifted their financing away from coal. We thus focus the analysis on bilateral funding agencies, which have been widely critiqued for being behind the majority of coal financing globally (Chen et al., 2021).

We confine our analysis to 2013–2017 due to data availability. More recent data sources are available, such as “Global Coal Public Finance Tracker” from Global Energy Monitor (2021a), “China's Global Energy Finance” from Boston University (Ma et al., 2020) and “Clean Energy Investment” from BloombergNEF (Climatescope, 2019). Yet these are unsuited to our research objective, which is to compare financing trends for both coal and renewables from bilateral government-affiliated organizations. Specifically, the former dataset focuses only on financing for coal, the second only provides data for China, and the latter does not include data on coal financing. Furthermore, we chose not to combine these datasets. This was to avoid mixing data reflecting different sources of financing (e.g. private and government-affiliated) and incompatible definitions of what constitutes “public financing” by each organization.

Based on the NRDC data, Table 5 shows the sample of 23 countries receiving international financing from a government-affiliated organization for both coal and renewable power generation projects. Countries are divided into two groups: (i) those receiving more financing for coal power, and (ii) those receiving more financing for renewables. 15 out of 23 countries received more foreign investments for CFPPs. These are primarily developing countries from Latin America (Brazil), Africa (Dominican Republic, Malawi, Mozambique, Zambia), Eastern Europe (Serbia, Ukraine), South Asia (Pakistan, Mongolia), and Southeast Asia (Indonesia, Vietnam). Together, these countries received more than \$41 billion for CFPPs over the study period; the bulk granted by

Table 4

Truth table.

Country	RE	GHG	BA	CP	NDC	DG	CR	NG	NI	EG	CFPP
Brazil	1	1	1	0	0	1	0	1	1	1	1
Colombia	0	0	1	0	0	0	1	1	1	1	1
Dominican Republic	0	1	0	0	0	0	0	1	1	1	1
Georgia	0	0	0	0	0	0	0	1	0	1	1
Greece	1	1	0	1	0	1	1	1	1	0	1
India	1	1	1	1	1	1	0	0	0	1	0
Indonesia	0	0	1	0	1	0	1	1	0	1	1
Israel	1	1	0	0	0	1	0	1	1	0	0
Kazakhstan	0	1	1	1	0	0	1	1	1	1	0
Kenya	0	0	0	0	0	0	0	0	0	1	0
Malawi	0	0	0	0	0	0	0	0	0	0	1
Mongolia	0	0	1	0	1	1	1	0	0	1	1
Morocco	1	0	0	0	0	1	0	1	0	1	0
Mozambique	0	0	1	0	0	0	0	1	0	0	1
Pakistan	0	0	1	0	1	0	0	1	0	0	1
Serbia	0	1	1	0	0	1	1	0	1	0	1
South Africa	1	0	1	1	1	1	1	0	1	0	0
Turkey	1	0	0	0	0	0	0	1	1	1	0
Ukraine	0	1	1	0	0	0	0	0	0	0	1
United Arab Emirates	1	0	0	0	0	0	0	1	1	0	0
Uzbekistan	0	1	1	0	0	0	0	1	0	0	1
Vietnam	1	0	1	0	0	0	0	1	0	1	1
Zambia	0	0	0	0	0	0	0	0	0	1	1

RE = Renewable energy policy; GHG = GHG emission target; BA = Bilateral agreement; NDC = Energy sector target in NDC; CP = Carbon price adoption; NI = National income; CR = Coal resources; NG = Access to natural gas; DG = Democratic functioning of the government; EG = Electrification growth; CFPP = More financing to coal-fired power plants (outcome).

Table 5

Overview of sample.

Recipient country	Total coal financing 2013–2017 (million USD)	Total renewables financing 2013–2017 (million USD)
Received more support for CFPPs		
Brazil	2769	1151
Colombia	300	9,750
Dominican Republic	632	40.5
Georgia	200	24
Greece	1017	182
Indonesia	17,884	1711
Malawi	600	10.4
Mongolia	1212	215
Mozambique	275	61.9
Pakistan	3473	1019
Serbia	1368	153
Ukraine	1300	64
Uzbekistan	165.6	110
Vietnam	9644	154
Zambia	515	58
Received more support for renewables		
India	1386	3663
Israel	96	814
Kazakhstan	210	282
Kenya	100	1912
Morocco	2050	5319
South Africa	1544	2109
Turkey	708.390	2011
United Arab Emirates	127.8	254

Data from Natural Resources Defense Council (NRDC, 2017).

government-affiliated institutions in China, Japan, and South Korea. This compares to just \$4 billion for renewables.

Application of final framework using coincidence analysis

Finally, we use coincidence analysis to mathematically identify the combination of factors that best explain why the international financing has flowed toward coal power more than renewables in the sampled countries. This is done for data collected for the period 2013–2017 for the influencing factors that: (i) were approved by the expert survey, and (ii) had data available for all countries. As explained above, this paper uses crisp (i.e. binary) values when constructing the truth table. The coincidence analysis was conducted with the CNA package available in the R program (version 4.0.3). We interpreted results using the robustness check to assess how much a particular model fits with others found in the same dataset, using various thresholds of accuracy and coverage.

Coincidence analysis is one of several configurational and comparative method (CCMs) developed for crisp-set data (Baumgartner & Ambühl, 2019). We selected it for two reasons: it can integrate both qualitative and quantitative data; and, compared to other CCMs such as qualitative comparative analysis, it is better suited to identifying correlations in data based on a bottom-up approach (Baumgartner & Ambühl, 2019; Haesebrouck, 2019; Whitaker et al., 2020). This means that individual factors are tested for sufficiency and necessity, then further tested in combinations of two, three, etc.

The following illustration, taken from Whitaker et al. (2020), shows how these methods can model the contribution of a complex condition in a data set to an outcome:

For $X \rightarrow Y$, if there is a fixed configuration of background factors Φ and change in the value of X is systematically linked with a change in the value of Y , then X is not a cause of Y if X does not make a difference to Y in any Φ .

The INUS theory of Mackie (1965), clarified later by Baumgartner (2008), is the most influential theory for defining the causality concept relevant for CCMs. An acronym for “Insufficient but Non-redundant (i.e., Necessary) part of an Unnecessary but Sufficient” condition, the INUS theory explains the sufficiency and necessity of a condition to arrive at the outcome. A simple illustration to explain INUS conditions is the correlation between rain and floods. A flood is not always caused by rain, since it can also be caused by a poor drainage system. However, the presence of rain in combination with other conditions—e.g. a poor drainage system—is sufficient for a flood. In this case, rain is an INUS condition. That is, it is unnecessary and insufficient without other conditions.

Coincidence analysis needs an optimal composition for the number of the cases and factors to get the best result. In the absence of literature explaining how this can be achieved, we followed the work of Marx and Dusa (2011) on qualitative comparative analysis as a benchmark to assure the validity of our coincidence analysis.⁴ Adopting this benchmark suggests that around 5–6 factors is the most optimal number for a sample of 23 cases. In our case, however, 10 influencing factors were examined in the coincidence analysis. This led to more than 900 combinations of factors as a final solution that can be seen in Supplementary material 5 (Table S5.1). The resulting model thus has a low diversity index and shows a massive ambiguity.

To minimize errors related to model specifications and select the best combination of factors from the atomic solution, we conducted a robustness check. From the theoretical foundation of coincidence analysis, the result shows no strong evidence for the final solution. Nonetheless, the final solution still can be selected based on the combination of the

Table 6
Final solutions.

Condition	Consistency	Coverage	Country
re*cp	0.917	0.733	Colombia, Dominican Republic, Georgia, Indonesia, <i>Kenya</i> , Malawi, Mongolia, Mozambique, Pakistan, Serbia, Ukraine, Uzbekistan, Zambia
BA*cp	1.000	0.677	Brazil, Colombia, Indonesia, Mongolia, Mozambique, Pakistan, Serbia, Ukraine, Uzbekistan, Vietnam
Solution	0.929	0.867	

Note: Kenya, shown in italics, indicates a deviant case, i.e., countries possessing the “solution” but receiving more international financing for renewables.

robustness check result and the background theory in the cause-effect hypothesis that we build from existing literature.⁵

Results

Interpretation of coincidence analysis

Table 6 shows the final solutions for the coincidence analysis. Three important factors (“re”, “cp”, “BA”) are identified. These emerged as the top three results in the robustness check table (see Supplementary material 5). Each of these: (i) does not largely overlap with all 900 combinations identified, (ii) has a sufficient consistency and coverage value, and (iii) is consistent with the background theory.

The coding appears in both different cases to indicate the presence (upper-case) or absence (lower-case) of factors in the truth table. Thus, “re” signifies the absence of strong policies supporting renewables, “BA” signifies the presence of a bilateral agreement for the coal sector, and “cp” shows absent or weak carbon pricing. The full script used for the coincidence analysis in R is provided in Supplementary material 6.

Two principal findings emerge from the final solution. First, countries were found to be more likely to receive more international financing for CFPPs if they had a bilateral agreement promoting the coal sector with investor countries (BA) combined with weak or missing adoption of carbon price (cp). Second, countries were also found to receive more financing for coal when characterized by the absence of strong supporting policies for renewable energy (re) combined with weak or missing carbon pricing (cp). Indeed, of the 15 countries receiving more funding for coal power than renewables, nine were found to satisfy the first combination of factors (BA*cp), while 12 met the second combination (re*cp).

Two important parameters in the solution are consistency and coverage. Consistency refers to the level of factual evidence supporting the identified relationship, while coverage defines how many cases are covered by the solution. The consistency and coverage of the final solutions are 0.929 and 0.867, respectively. This means the solutions are supported by a high level of evidence (93%) and coverage (87%). Indeed, we find only one deviant case (Kenya), which appears in the first solution. This deviant case refers to countries that met the criteria of the solution, but received more financing for renewables, which is the opposite of the expected outcome.

This finding about the influence of carbon pricing somewhat confirms the “polluter’s haven” expectation. But the solutions also point to a novel finding—that the presence of a bilateral commitment with a recipient country to develop the coal industry is a substantial driver of international financing in CFPPs. Having a bilateral agreement that promotes coal-industry development is not sufficient to explain the outcome; however, this factor in combination with weak or absent carbon price was observed in 60% of the sampled countries that receive more financing for coal than renewables.

⁵ We are grateful to Professor Michael Baumgartner at University of Bergen for explaining this point.

⁴ However, this benchmark has some limitations. The random data used by the authors is one problem. The overly low sample sizes and the informativeness of the output—which they do not discuss—is another. In addition, this research is also not very explicit about the fact that their benchmarks are mere rules of thumb that work on averages.

Table 7
Raw data factors from solution and CFPP development status.

Country	Raw data of influential factors from CNA solution			CFPPs financing (million USD)	Renewables financing (million USD)	CFPP status (as of January 2018)	
	Renewable energy policy (RE)	Bilateral agreement (BA)	Carbon price adoption (CP)			Pre-construction & construction (MW)	Operating (MW)
Brazil	High	Yes	Low	2769	1151	2906	4355
Colombia	Medium	Yes	Low	300	9,750	1150	1393
Dominican Republic	Medium	No	Low	632	40.5	770	305
Georgia	NA	No	Low	200	24	300	0
Greece	High	No	High	1017	182	1110	4375
India	High	Yes	High	1386	3663	131,359	214,910
Indonesia	Medium	Yes	Medium	17,884	1711	37,905	28,584
Israel	High	No	Medium	96	814	0	4900
Kazakhstan	Medium	Yes	High	210	282	636	12,000
Kenya	Medium	No	Low	100	1912	1050	0
Malawi	Medium	No	Low	600	10.4	2820	0
Mongolia	Low	Yes	Low	1212	215	7665	706
Morocco	High	No	Medium	2050	5319	1736	2585
Mozambique	Low	Yes	Low	275	61.9	2440	0
Pakistan	Medium	Yes	Low	3473	1019	12,385	2450
Serbia	Medium	Yes	Low	1368	153	2850	4405
South Africa	High	Yes	High	1544	2109	11,892	42,101
Turkey	High	No	Medium	708	2011	42,890	18,469
Ukraine	Medium	Yes	Medium	1300	64	1320	23,259
United Arab Emirates	High	No	Medium	127.8	254	5400	0
Uzbekistan	Medium	Yes	Medium	165.6	110	0	2522
Vietnam	High	Yes	Low	9644	154	46,525	14,971
Zambia	Medium	No	Low	515	58	300	330

Data sources: Renewable energy policy (World Bank Group, 2017); bilateral agreements (own data); carbon price adoption (World Bank Group, 2017); renewables financing (NRDC, 2017), CFPPs financing (NRDC, 2017); CFPP construction and operation (Global Energy Monitor's Global Coal Plant Tracker, 2020).

Discussion of findings

This section uses qualitative evidence from literature to explore the influence of the factors identified through the coincidence analysis. In addition to the study period (i.e. the timeframe for which we ran the coincidence analysis [2013–2017]), we also source evidence after this period. We focus on four countries. Three of these—Indonesia, Mongolia, and Pakistan—fit with both solutions (BA*cp and re*cp), and Vietnam fits with just one (BA*cp). In selecting these four, we also noted that, at the end of the study period in early 2018, each country's CFPP pipeline was the largest of all listed in the final solution (see Table 7). Focusing on these four countries thus allows us to explore how the identified factors have influenced the development and construction of CFPPs in major global hotspots.

Indonesia (re*cp + BA*cp)

Indonesia's fleet of CFPPs has expanded rapidly since around 2010, with the government striving to provide cheap electricity to the growing population and economy (Kurniawan et al., 2020; Ordóñez et al., 2021a). In 2020 it had around 34 GW of installed capacity, and added around 22 GW from 2010 to 2020 (Global Energy Monitor, 2021a). During the study period (2013–2017), international financing received from government-affiliated organizations for coal power totaled \$17 billion, compared to \$1 billion for renewables (NRDC, 2017). Qualitative evidence supports the finding that this situation is largely explained by the absence of carbon price (cp) and strong policy support for renewables (re) and by the presence of bilateral agreements promoting coal industry development (BA).

Beginning with policy, in 2017 the Indonesian government set a goal to raise the share of renewables to 23% of overall energy consumption by 2025, and to 31% by 2050 (PWC, 2018). Yet policy support for coal has been stronger. In 2019, a national goal was set to add 21 GW of CFPP capacity over the next 10 years, allowing coal to make up around 50% of electricity generation in 2025 (PLN, 2019). Restrictive rules and policies have hampered achievement of the conservative targets for renewables, creating conditions favoring the production and distribution of coal power (ADB, 2020).

A noteworthy example is pricing regulations. Solar, wind, and geothermal are not expected to achieve cost parity with coal until about 2030 (Arinaldo & Adiatma, 2019). But despite the need for protective pricing arrangements until this occurs, in 2017 the Ministry of Energy and Mineral Resources ruled that renewable electricity, to qualify for purchasing by the national and state-owned power utility PLN, must be procured at 85%–100% of the cost of PLN's average output. Consequently, renewables must compete with the economics of PLN's vast baseload of coal electricity, where costs are further depressed by subsidies for both CFPPs and extraction (Bridle et al., 2018; Kurniawan et al., 2020). Many independent power producers who considered developing renewable generation projects in Indonesia's rapidly growing electricity market were disincentivized by such pricing rules and chose coal instead (ADB, 2020).

The absence of carbon price exacerbates this situation, allowing CFPP operators to produce cheap, subsidized electricity without having to internalize environmental externalities, such as carbon emissions and air pollution (Ordóñez et al., 2021b; Tritto, 2021). In 2021, the Indonesian state is reported to be investigating the introduction of a carbon tax (Thomas, 2021). Until now, however, this has been hampered by political barriers, including industry opposition on the basis of fears around compromised competitiveness and economic growth (Dyarto & Setyawan, 2020).

Bilateral agreements forged with investor countries also appear to have driven international financing in CFPPs. One partnership during the study period was forged with Japan in 2015 (Ministry of Foreign Affairs of Japan, 2015). This joint statement affirmed Japan's commitment to helping Indonesia build high-efficiency CFPPs as part of a broader ambition to export other infrastructure packages, such as shipping ports and mass rapid transport. It also pledges for Japan to lever public funds to help Indonesia achieve its CFPP expansion program. A mixture of public and private financing has flowed from Japan to Indonesia under this arrangement. For instance, Japanese public finance institutions provided more than \$3 billion for the 2 GW Java Batang project in 2017 (Han & Jack, 2017; Woods, 2017), and in 2020 JICA was reported to be considering investing \$1.8 billion in the 1 GW Indramayu power station (Global Energy Monitor, 2021a).

Strategic partnerships with China have also propelled the influx of Chinese investments and financing for new CFPP constructions in Indonesia. This has occurred principally under the One Belt One Road initiative since 2013 (Tritto, 2021). Through this initiative, state-owned companies and banks (e.g. China Development Bank) have provided low-interest loans, cheap technologies and labor as well as engineering, construction, and operation (Mori, 2020), with China's presence in CFPP development surpassing Japan's historical lead. Chinese investments amounted to \$13.4 billion over 2000–2016, targeting the development of several big CFPP projects, such as the Jawa 9–10 (2 GW), Indramayu (1 GW), and Celukan Bawang (660 MW) (Endcoal.org, 2020; Tritto, 2021). Funding has continued beyond the study period (Gallagher et al., 2021), a recent example being the construction of the Bangko Tengah power plant (1.2 GW) in 2018 (Shearer et al., 2020).

Bilateral agreements with China and Japan have also encouraged the development of coal mines in Indonesia. In addition to the opportunities provided by CFPP construction for Chinese and Japanese technology manufacturers to profit from emerging markets in Asia (Trencher, Downie et al., 2020; Yao et al., 2018), upstream investments have also aimed to secure prioritized access to Indonesia's coal reserves. With this coal subsequently imported and consumed on home soil by China and Japan, bilateral developments thereby strengthen the security of fuel supply chains for Indonesia's foreign partners (Mori, 2018; Trencher, Downie et al., 2020).

*Pakistan ($re^*cp + BA^*cp$)*

Qualitative evidence from Pakistan also supports the results of the coincidence analysis. Here too we find that the existence of bilateral agreements in the coal industry (BA), and the absence of strong policy support for renewable energy (re) and carbon pricing (cp), have largely contributed to driving investments and financing for coal power. Pakistan's CFPP installed capacity was around 5 GW in 2020 (Global Energy Monitor, 2021a), 60% of which was built in 2016–2018. During the study period, financing received from foreign-government-affiliated organizations amounted to \$3 billion, compared to around \$1 billion for renewables (NRDC, 2017).

We begin with the absence of strong supporting policies for renewables. In 2019, the Pakistani government fixed the objective of expanding renewables over the period 2018–2040, aiming to add 31 GW from hydro and 15 GW from other sources. However, this plan also privileges coal, aiming to add 27 GW. The development of renewables, meanwhile, faces numerous challenges (Zafar et al., 2018). The first problem is cost competitiveness: while renewable power generation is expensive due to the absence of local manufacturers and the need to import equipment, electricity from fossil fuels (e.g. crude oil) is highly subsidized and bought by the government above market price (Ali et al., 2021; Hassan et al., 2021). Second, lack of coordination across government departments is hampering the development of strong incentive mechanisms for renewables (Khan et al., 2020). Third, time-consuming and complex permit application procedures also dissuade investments in renewables (Zafar et al., 2018).

Pakistan, like Indonesia, also lacks a carbon price scheme. Although several national policy documents could provide a legal basis for such regulation, the degree of support among politicians, industry, and the general public is relatively low due to perceptions that such mechanisms would burden taxpayers (UNFCCC, 2019).

Evidence also suggests that bilateral agreements emphasizing the development of coal power and mining have been an important driver of CFPP financing. An important bilateral arrangement with China occurred under the 2015 China–Pakistan Economic Corridor (CPEC) (Ministry of Planning Development, and Reform, 2017). Formed as part of China's larger One Belt One Road initiative, these two commitments stress coal extraction as a key area for bilateral cooperation, directing both sides to optimize the use of Pakistan's coal reserves.

China benefits by receiving access to coal resources for import, while Pakistan is incentivized to facilitate the Chinese-led construction of CFPPs to stabilize its limited electricity supply, which has historically suffered from chronic blackouts (Umbach & Yu, 2010).

These arrangements have propelled funding into the development of coal mines and CFPPs, during the study period and more recently. For example, the Chinese government and banks have funded the CPEC initiative with \$27.6 billion in electricity and infrastructure projects in 2015–2021, much of which has gone into energy (and specifically coal) projects (Umbach & Yu, 2010). Indeed, since 2018, Chinese funders and plant developers have been involved in about half of the newly constructed CFPPs in Pakistan (Shearer et al., 2018).

*Mongolia ($re^*cp + BA^*cp$)*

Mongolia, like the other cases examined, has looked to coal to expand its rapidly growing electricity supply. This is incentivized by the availability of rich domestic reserves (Hans et al., 2020) and by the need to supply affordable power in a poor economy, where some 10% of the population lack access (OECD, 2019). Although the installed capacity of CFPPs is only 0.8 GW as of 2020, Mongolia has the fourth largest pipeline in the world after Indonesia, Vietnam and Pakistan (Global Energy Monitor, 2021a). During the study period of 2013–2017, financing for coal power from foreign-government-affiliated organizations amounted to around \$1 billion; renewables received less than a quarter of this (NRDC, 2017). With only 0.2 GW of CFPP added in 2016–2019, Mongolia's pipeline continues to grow, with some 7 GW of projects currently under development (Global Energy Monitor, 2021a). Qualitative evidence also suggests, as with Indonesia and Pakistan, that the absence of strong policy support for renewable energy (re) and carbon price (cp), coupled with the presence of bilateral agreements in the coal industry (BA), have stimulated this historical and ongoing coal power financing.

Like Indonesia and Pakistan, Mongolia also lacks robust policies to support renewable energy. In 2015, issuing its Green Development Policy, the Mongolian government set the goal of increasing the capacity of renewable power production by 20% by 2023 and by 30% by 2030 (IRENA, 2016). Yet subsequent policies to achieve this have been largely insufficient and poorly implemented, with government institutions mostly supporting coal power development (Ana-Maria Seman, 2017) due to its abundance and consequent low cost (Hans et al., 2020). For instance, although Mongolia has a FIT scheme, its 10-year duration has been insufficient to attract large investments to renewables (IRENA, 2016).

Case-based evidence also suggests the potential influence of bilateral agreements on driving CFPP development. Formal bilateral ties with Japan were strengthened in 2015. This agreement emphasized the importance of developing Mongolia's mineral resources, including coal deposits (Sutton, 2015). With Japanese mining companies later increasing investments in extraction (NEDO, 2016), Japanese financial institutions such as Japan Bank for International Cooperation (JBIC) and the Nippon Export and Investment Insurance (NEXI) have made important financial contributions to the coal sector alongside South Korea and China (Jiaqiao et al., 2018). For example, around \$300 million of investments from Japan were directed at the Tavan Tolgoi Coal Power Plant (600 MW) that planned to build in 2020 (Ana-Maria Seman, 2017). More recently, bilateral agreements continue to drive investments in coal. China's One Belt One Road initiative has forged formal agreements with Mongolia that also target coal resource development (Ulagan, 2021). Power-sector investments unfolding through this initiative since 2017 have also focused on coal (Gu et al., 2020).

*Vietnam (BA^*cp)*

Compared to the other countries, Vietnam's situation is unique. The solution shows that despite having strong policy support for renewable energy, it received more international financing for CFPPs due to the

combination of bilateral agreements (BA) and weak carbon-price adoption (cp). As of 2020, Vietnam had an installed capacity of CFPPs that reached 19 GW. This fleet has grown rapidly: around 70% of new constructions were concentrated between 2013 and 2018 (Global Energy Monitor, 2021a). international financing in CFPPs from government-affiliated organizations over the study period reached \$9 billion, whereas renewables received only around \$0.1 billion (NRDC, 2017). Evidence shows that these factors are still relevant, as Vietnam's CFPP fleet continues to grow at an unprecedented pace (Dorband et al., 2020; Gallagher et al., 2021). As of 2020, it possessed around 30 GW of new CFPP development in the pipeline (Global Energy Monitor, 2021b).

Beginning with bilateral agreements, Vietnam and Japan forged a strategic partnership in 2017 (The Ministry of Foreign Affairs of Japan, 2017). In the broader goal of pledging to help Vietnam develop its economy via the export of Japanese infrastructure project, this partnership stated a goal for Japan to provide engineering expertise in the energy sector. More specifically, it outlined an explicit goal for Japan to construct three CFPPs. The realization of this agreement can be seen from large energy deals that occurred in 2019. One notable project financed by Japan involved a loan of around \$2 billion from JBIC to the Quang Tri 2 Power Plant (1.2 GW) and Vung Ang 2 Power plant (1.2 GW).

Regulations to discourage carbon-intensive financing in the power sector, such as carbon pricing, were absent during the study period. In late 2020, Vietnam's National Assembly voted to introduce an emissions trading scheme, with implementation planned after 2022. Like in Indonesia, the historical delay in introducing such carbon pricing can be explained by the resistance of government and industry based on fears of raising electricity tariffs, increasing prices for consumers (Dorband et al., 2020), and hampering economic growth (UNDP, 2018).

Conclusion

This article analyzed the factors that encourage international financing toward either coal or renewables in countries receiving international financial support for power generation sectors. The results indicate that countries characterized by the combined absence of strong policies supporting renewables and carbon price adoption are most likely to receive more financing for coal power. While this is not a surprising finding, our study also revealed the novel insight that two other factors—the presence of bilateral agreements promoting coal-industry development, and the absence of carbon price adoption—can also explain why some countries receive greater public financing from foreign institutions toward coal than renewables. The results contribute to a broader understanding of how international support for power-plant developments across the world is influenced not only by the motivations of investor countries, but equally by the factors in recipient countries. This has been underemphasized in literature to date (Chen et al., 2020a; Gallagher, 2018; Gallagher et al., 2018).

Investments in CFPPs continue to decline (IEA, 2020), and some investors have started to pull out of planned projects (Manych et al., 2021; Uddin et al., 2019). Rather than merely trying to curb the flow of international financing to coal power, recipient countries could focus efforts on improving their investment environment to be more friendly to renewable energy. Many scholars have discussed the importance of policy and regulation to attract international investment to renewable energy. Yet our findings suggest that negotiating bilateral or multilateral agreements that affect infrastructure and energy resource development is a crucial instrument where commitments to decarbonized electricity generation need to occur.

In parallel, the importance of carbon pricing regulation should be accounted for. As suggested by the prevalence of carbon pricing schemes in the sampled countries receiving more financing for renewables, the political and socio-economic feasibility to introduce such regulations in the many countries still building CFPPs may be a crucial determinant of the carbon intensity of future financing in the power sector.

Several methodological limitations deserve emphasis. Our analysis was limited to the specific period for which data was available, since we required financing amounts made for both coal and renewables from government-affiliated institutions, which make up the bulk of international power-sector investments. Other limitations stemmed from difficulties in operationalizing and dichotomizing certain variables that were accepted by the expert survey, but for which data was not fully available for all sampled countries. These include “renewable electricity price,” “CFPP electricity price,” and “environmental awareness.”

Future research could address these limitations to deepen understanding on a broader range of more recent trends affecting the carbon intensity of power-sector investments globally. Questions also remain about the relative influence of, and interactions or interdependencies between, the factors identified in the coincidence analysis solutions. Future studies could use a mixture of quantitative data and qualitative case studies using various sources (e.g. interviews and document analysis) to further explore the nature of factors that affect international financing and investments in coal and renewables.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors express gratitude to Professor Michael Baumgartner at University of Bergen for valuable guidance in applying the coincidence analysis method. We also thank all the organizations and respondents who kindly completed our survey as well as Professor Purnomo Yugianoro at Purnomo Yugianoro Center (PYC) for insightful feedback, and Dr. Jan Steckel at Mercator Research Institute on Global Commons and Climate Change (MCC) in Berlin for valuable advice when planning this study. We also acknowledge support received from the German-Japanese University Network (HeKKSaGOn).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.esd.2021.12.004>.

References

- ADB (2020). *Energy sector assessment, strategy, and road map: Indonesia, country sector and thematic assessments*.
- Ali, Z., et al. (2021). Original article a comparison of energy policies of Pakistan and their impact on bioenergy development. *Sustainable Energy Technologies and Assessments*, 46(April), 101246. <https://doi.org/10.1016/j.seta.2021.101246> Elsevier Ltd.
- Alova, G., Trotter, P. A., & Money, A. (2021). A machine-learning approach to predicting Africa's electricity mix based on planned power plants and their chances of success. *Nature Energy*, 6(2), 158–166. <https://doi.org/10.1038/s41560-020-00755-9> Springer US.
- Arinaldo, D., & Adiatma, J. C. (2019). Indonesia's coal dynamics: Toward a just energy transition. Available at: http://iesr.or.id/wp-content/uploads/2019/08/Indonesias-Coal-Dynamics_Toward-a-Just-Energy-Transition.pdf.
- Baumgartner, M. (2008). Regularity theories reassessed. *Philosophia*, 36(3), 327–354. <https://doi.org/10.1007/s11406-007-9114-4>.
- Baumgartner, M., & Ambühl, M. (2019). *cna: An R package for configurational causal inference and modeling*, 1–36.
- Bertram, C., et al. (2015). Carbon lock-in through capital stock inertia associated with weak near-term climate policies. *Technological Forecasting and Social Change*, 90(PA), 62–72. <https://doi.org/10.1016/j.techfore.2013.10.001> Elsevier Inc.
- Best, R. (2017). Switching towards coal or renewable energy? The effects of financial capital on energy transitions. *Energy Economics*, 63, 75–83. <https://doi.org/10.1016/j.eneco.2017.01.019> Elsevier B.V.
- Blondeel, M., Van de Graaf, T., & Haesebrouck, T. (2020). Moving beyond coal: Exploring and explaining the Powering Past Coal Alliance. *Energy Research and Social Change*, 59 (September 2019), 101304. <https://doi.org/10.1016/j.erss.2019.101304> Elsevier.
- Bohlmann, H. R., et al. (2019). Regional employment and economic growth effects of South Africa's transition to low-carbon energy supply mix. *Energy Policy*, 128(February), 830–837. <https://doi.org/10.1016/j.enpol.2019.01.065> Elsevier Ltd.

- Böhringer, C., Rosendahl, K. E., & Storrøsten, H. B. (2017). Robust policies to mitigate carbon leakage. *Journal of Public Economics*, 149, 35–46. <https://doi.org/10.1016/j.jpubeco.2017.03.006> Elsevier B.V.
- Bridle, R., et al. (2018). Missing the 23 per cent target: Roadblocks to the development of renewable energy in Indonesia. Available at: www.iisd.org/gsi.
- Carbon Brief (2015). *Paris 2015: Tracking country climate pledges* [Carbon Brief]. Carbon Brief Available at: <https://www.carbonbrief.org/paris-2015-tracking-country-climate-pledges> (Accessed: 2 November 2020).
- Chapman, A., & Itaoka, K. (2018). Curiosity, economic and environmental reasoning: Public perceptions of liberalization and renewable energy transition in Japan. *Energy Research and Social Science*, 37(October 2017), 102–110. <https://doi.org/10.1016/j.erss.2017.09.026> Elsevier.
- Chen, H., et al. (2016). *Carbon trap: How international coal finance undermines the Paris Agreement*.
- Chen, X., Gallagher, K. P., & Mauzerall, D. L. (2020a). Article Chinese overseas development financing of elec
- Chen, X., Gallagher, K. P., & Mauzerall, D. L. (2020b). Article Chinese overseas development financing of elec
- Chen, X., et al. (2021). Financing carbon lock-in in developing countries: Bilateral financing for power generation technologies from China, Japan, and the United States. *Applied Energy*, 300, 117318. <https://doi.org/10.1016/j.apenergy.2021.117318> Elsevier Ltd.
- CLIMATESCOPE (2019). Emerging markets cross-border clean energy investment. Available at: <https://global-climatescope.org/clean-energy-investments>.
- Cole, M. A. (2004). Trade, the pollution haven hypothesis and the environmental Kuznets curve: Examining the linkages. *Ecological Economics*, 48(1), 71–81. <https://doi.org/10.1016/j.ecolecon.2003.09.007>.
- Diluiso, F., Walk, P., Many, N., et al. (2021). Coal transitions-Part 2: A systematic map and review of case study learnings from regional, national, and local phase-out experiences. *Environmental Research Letters*, 16, Article 113003. <https://doi.org/10.1088/1748-9326/ac1b58>.
- Dorband, I. I., Jakob, M., & Steckel, J. C. (2020). Unraveling the political economy of coal: Insights from Vietnam. *Energy Policy*, 147(August), 111860. <https://doi.org/10.1016/j.enpol.2020.111860> Elsevier Ltd.
- Dorsey, J. (2019). *Waiting for the courts: Effects of policy uncertainty on pollution and investment, environmental and resource economics*. Netherlands: Springer. <https://doi.org/10.1007/s10640-019-00375-2>.
- Dyarto, R., & Setyawan, D. (2020). Understanding the political challenges of introducing a carbon tax in Indonesia. *International Journal of Environmental Science and Technology*. <https://doi.org/10.1007/s13762-020-02925-4> Springer Berlin Heidelberg, (0123456789).
- Edenhofer, O., et al. (2018). Reports of coal's terminal decline may be exaggerated. *Environmental Research Letters*, 13(2). <https://doi.org/10.1088/1748-9326/aaa3a2>.
- Endcoal.org (2020). Global coal plant tracker. Available at: https://docs.google.com/spreadsheets/d/1W-gobEQugqTR_PP0icZjCrdaR-vYkJOdztsSsCjXuKw/edit#gid=0.
- Erickson, P., et al. (2015). Assessing carbon lock-in. *Environmental Research Letters*, 10(8). <https://doi.org/10.1088/1748-9326/10/8/084023> IOP Publishing.
- Freedom of House (2018). *FWI all data FWI 2013–2018*.
- Gabriel, C. A. (2016). What is challenging renewable energy entrepreneurs in developing countries? *Renewable and Sustainable Energy Reviews*, 64, 362–371. <https://doi.org/10.1016/j.rser.2016.06.025> Elsevier.
- Gallagher, K. P. (2018). China's global energy finance: Poised to lead. *Energy Research and Social Science*, 35(February), 15–16. <https://doi.org/10.1016/j.erss.2018.01.001> Elsevier.
- Gallagher, K. P., et al. (2018). Energizing development finance? The benefits and risks of China's development finance in the global energy sector. *Energy Policy*, 122(August), 313–321. <https://doi.org/10.1016/j.enpol.2018.06.009> Elsevier Ltd.
- Gallagher, K. S., & Qi, Q. (2018). Policies governing China's overseas development finance implications for climate change. Available at: https://sites.tufts.edu/cierp/files/2018/03/CPL_ChinaOverseasDev.pdf.
- Gallagher, K. S., et al. (2021). Banking on coal? Drivers of demand for Chinese overseas investments in coal in Bangladesh, India, Indonesia and Vietnam. *Energy Research and Social Science*, 71, 101827. <https://doi.org/10.1016/j.erss.2020.101827> Elsevier Ltd.
- Global Energy Monitor (2021a). *Boom and bust 2021*. <https://doi.org/10.1016/j.jacr.2018.08.035>.
- Global Energy Monitor (2021b). *New coal plants by country*. Available at: <https://endcoal.org/global-coal-plant-tracker/>.
- Global Energy Monitor's Global Coal Plant Tracker (2020). *Coal plants by country*. Global Energy Monitor's Global Coal Plant Tracker Available at: <http://endcoal.org/wp-content/uploads/2017/03/Jan-2017-Proposed-by-country-units.pdf>.
- Gu, B., Novikov, V., & Simonett, O. (2020). *China–Mongolia–Russia economic corridor*.
- Guzel, A. E., & Okumus, I. (2020). Revisiting the pollution haven hypothesis in ASEAN-5 countries: New insights from panel data analysis. *Environmental Science and Pollution Research*, 27(15), 18157–18167. <https://doi.org/10.1007/s11356-020-08317-y>.
- Haesebrouck, T. (2019). Who follows whom? A coincidence analysis of military action, public opinion and threats. *Journal of Peace Research*, 56(6), 753–766. <https://doi.org/10.1177/0022343319854787>.
- Han, C., & Jack, S. (2017). *Power shift: Shifting G20 international public finance from coal to renewables*. National Resources Defence Council (NRDC) Available at: <https://www.nrdc.org/resources/power-shift-g20-international-public-finance-from-coal-to-renewables>.
- Hans, F., et al. (2020). The Mongolian electricity sector in the context of international climate mitigation efforts. Available at: https://newclimate.org/wp-content/uploads/2020/03/Decarbonization_Pathways_Mongolia.pdf.
- Hassan, A., et al. (2021). Monetization of the environmental damage caused by fossil fuels. *Environmental Science and Pollution Research*, 28(17), 21204–21211. <https://doi.org/10.1007/s11356-020-12205-w>.
- Ialnazov, D., & Keeley, A. (2020). Motivations, enabling factors and barriers to the energy transition in Indonesia and Vietnam. *IOP Conference Series: Earth and Environmental Science*, 505, Article 012044. <https://doi.org/10.1088/1755-1315/505/1/012044>.
- IEA (2020). Data and statistics. Available at: <https://www.iea.org/data-and-statistics?country=WORLD&fuel=Electricity&heat&indicator=Electricitygenerationbysource%0A%0A> (Accessed: 17 February 2020).
- IEA (2020). World energy investment 2020. *World energy investment 2020*. <https://doi.org/10.1787/6f552938-en>.
- IEEFA (2021). *Over 100 and counting*. Institute for Energy Economics & Financial Analysis Available at: <https://ieefa.org/finance-exiting-coal/> (Accessed: 5 February 2021).
- IPCC (2018). Summary for policymakers. *Global warming of 1.5°C. An IPCC special report on the impacts of global warming of 1.5°C above pre-industrial levels and related global greenhouse gas emission pathways, in the context of strengthening the global response to, World Meteorological Organization, Geneva, Switzerland* Available at: https://www.ipcc.ch/site/assets/uploads/sites/2/2019/05/SR15_SPM_version_report_LR.pdf.
- IRENA (2016). *Renewables readiness assessment Mongolia*. Mongolia: Renewables Readiness Assessment.
- Janzwood, A. (2020). Explaining variation in oil sands pipeline projects. *Canadian Journal of Political Science*, 1–20. <https://doi.org/10.1017/S0008423920000190>.
- Jewell, J., et al. (2019). Prospects for powering past coal. *Nature Climate Change*, 9(8), 592–597. <https://doi.org/10.1038/s41558-019-0509-6> Springer US.
- Jiaqiao, L., Momoi, T., & Lee, J. (2018). Coal power sector in China, Japan and South Korea. Available at: <https://www.kikonet.org/wp/wp-content/uploads/2018/12/coal-report-in-CH-JP-KR-2018.pdf>.
- Keeley, A. R., & Matsumoto, K. (2018a). Investors' perspective on determinants of foreign direct investment in wind and solar energy in developing economies – Review and expert opinions. *Journal of Cleaner Production*, 179, 132–142. <https://doi.org/10.1016/j.jclepro.2017.12.154> Elsevier Ltd.
- Keeley, A. R., & Matsumoto, K. (2018b). Relative significance of determinants of foreign direct investment in wind and solar energy in developing countries – AHP analysis. *Energy Policy*, 123(September), 337–348. <https://doi.org/10.1016/j.enpol.2018.08.055> Elsevier Ltd.
- Khan, M. I., Khan, I. A., & Chang, Y. C. (2020). An overview of global renewable energy trends and current practices i Pakistan - A perspective of policy implications. *Journal of Renewable and Sustainable Energy*, 12(5). <https://doi.org/10.1063/5.0005906> AIP Publishing LLC.
- Koplietz, S. N., et al. (2017). Burden of disease from rising coal-fired power plant emissions in Southeast Asia. *Environmental Science and Technology*, 51(3), 1467–1476. <https://doi.org/10.1021/acs.est.6b03731>.
- Koppenborg, F. (2016). Japanese climate-related aid to South East Asia: Furthering “weak”, “medium” or “strong” sustainability? *Florentine Koppenborg. Asien*, 138(S), 31–51.
- Kox, H. L. M., & Rojas-Romagos, H. (2020). How trade and investment agreements affect bilateral foreign direct investment: Results from a structural gravity model. *The World Economy*, 43(12), 3203–3242. <https://doi.org/10.1111/twec.13002>.
- Kurniawan, R., et al. (2020). Understanding the multi-faceted drivers of increasing coal consumption in Indonesia. *Energies*, 13(14), 3660. <https://doi.org/10.3390/en13143660>.
- Lamb, W. F., & Minx, J. C. (2020). The political economy of national climate policy: Architectures of constraint and a typology of countries. *Energy Research and Social Science*, 64(January), 101429. <https://doi.org/10.1016/j.erss.2020.101429> Elsevier.
- Lin, B., & Bega, F. (2021). China's Belt & Road Initiative coal power cooperation: Transitioning toward low-carbon development. *Energy Policy*, 156(June), 112438. <https://doi.org/10.1016/j.enpol.2021.112438> Elsevier Ltd.
- Ma, X., Gallagher, K. P., & Guo, Y. (2020). *China's global energy finance 2019*. Global Development Policy Center Available at: <https://www.bu.edu/cgef/#/all/EnergySource/Coal> (Accessed: 4 July 2021).
- Mac Kinnon, M. A., Brouwer, J., & Samuelsen, S. (2018). The role of natural gas and its infrastructure in mitigating greenhouse gas emissions, improving regional air quality, and renewable resource integration. *Progress in Energy and Combustion Science*, 64, 62–92. <https://doi.org/10.1016/j.peccs.2017.10.002> Elsevier Ltd.
- Mackie, J. L. (1965). Causes and conditions. *American Philosophical Quarterly*, 2(4), 245–264 (North American Philosophical Publications, University of Illinois Press). Available at: <http://www.jstor.org/stable/20009173>.
- Many, N., Steckel, J. C., & Jakob, M. (2021). Finance-based accounting of coal emissions. *Environmental Research Letters*, 16(4). <https://doi.org/10.1088/1748-9326/abd972>.
- Marx, A., & Dusa, A. (2011). Crisp-set qualitative comparative analysis (csQCA), contradictions and consistency benchmarks for model specification. *Methodological Innovations Online*, 6(2), 103–148. <https://doi.org/10.4256/mio.2010.0037>.
- Ministry of Foreign Affairs of Japan (2015). Japan - Indonesia joint statement towards further strengthening of the strategic partnership underpinned by sea and democracy. Japan. Available at: <https://www.mofa.go.jp/files/100002843.pdf>.
- Ministry of Planning Development & Reform, G. of P (2017). *Long term plan for China - Pakistan economic corridor (2017–2030)*. Ministry of Planning Development & Reform, Government of Pakistan Available at: <http://pc.gov.pk/uploads/cpec/LTP.pdf>.
- Mori, A. (2018). Chapter 10 impact of the China-induced coal boom in Indonesia: From a resource governance perspective. *China's climate and energy policy*.
- Mori, A. (2020). Foreign actors, faster transitions? Co-evolution of complementarities, perspectives and sociotechnical systems in the case of Indonesia's electricity supply system. *Energy Research and Social Science*, 69(July 2019), 101594. <https://doi.org/10.1016/j.erss.2020.101594> Elsevier.
- Nace, T. (2018). A coal phase-out pathway for 1.5°C. Available at: <https://storage.googleapis.com/planet4-international-stateless/2018/10/7df76ee5-coalpathway-final.pdf>.
- NEDO (2016). Japan-Mongolia joint project to start at Tavan Tolgoi Coal Mine in Mongolia. Available at: https://www.nedo.go.jp/english/news/AA5en_100039.html.
- NRDC (2017). *NRDC-consolidated-coal-renewable-database-2017*.
- Oberschelp, C., et al. (2019). Global emission hotspots of coal power generation. *Nature Sustainability*, 2(2), 113–121. <https://doi.org/10.1038/s41893-019-0221-6> Springer US.

- OECD (2019). Sustainable infrastructure for low-carbon development in Central Asia and the Caucasus. Available at: <https://www.oecd.org/publications/sustainable-infrastructure-for-low-carbon-development-in-central-asia-and-the-caucasus-d1aa6ae9-en.htm>.
- Ordóñez, J.A., et al. (2021a). Coal, power and coal-powered politics in Indonesia. *Environmental Science and Policy*, 123(June 2020), 44–57. <https://doi.org/10.1016/j.envsci.2021.05.007> Elsevier Ltd.
- Ordóñez, J.A., et al. (2021b). Coal, power and coal-powered politics in Indonesia. *Environmental Science & Policy*, 123(April), 44–57. <https://doi.org/10.1016/j.envsci.2021.05.007> Elsevier Ltd.
- Peabody (2017). *2017 corporate social responsibility report*.
- Pelham, B. W. (2009). Awareness, opinions about global warming vary worldwide. Gallup Available at: <http://www.gallup.com/poll/117772/Awareness-Opinions-Global-Warming-Vary-Worldwide.aspx?version=print> (Accessed: 10 August 2020).
- PT. PLN (Persero) (2019). Electric power supply business plan (2019–2028). Available at: http://gatrik.esdm.go.id/assets/uploads/download_index/files/5b16d-kepmen-esdm-no.-39-k-20-mem-2019-tentang-pengesahan-ruptl-pt-pln-2019-2028.pdf.
- PWC (2018). *Power in Indonesia*.
- Ragosa, G., & Warren, P. (2019). Unpacking the determinants of cross-border private investment in renewable energy in developing countries. *Journal of Cleaner Production*, 235, 854–865. <https://doi.org/10.1016/j.jclepro.2019.06.166> Elsevier Ltd.
- Rasheed, R., et al. (2021). Life cycle assessment of a cleaner supercritical coal-fired power plant. *Journal of Cleaner Production*, 279, 123869. <https://doi.org/10.1016/j.jclepro.2020.123869> Elsevier Ltd.
- Rentier, G., Lelieveldt, H., & Kramer, G. J. (2019). Varieties of coal-fired power phase-out across Europe. *Energy Policy*, 132(June), 620–632. <https://doi.org/10.1016/j.enpol.2019.05.042> Elsevier Ltd.
- Schmid, B., & Bornemann, B. (2019). What political settings promote renewable energy investments by energy utilities?—A qualitative comparative analysis in Swiss Cantons. *European Policy Analysis*, 5(2), 232–265. <https://doi.org/10.1002/epa2.1055>.
- Seman, Ana-Maria (2017). *Mongolia's energy sector*. CEE Bankwatch Network Available at: <https://bankwatch.org/sites/default/files/Mongolia-energy-sector-web.pdf>.
- Sen, S., & Ganguly, S. (2017). Opportunities, barriers and issues with renewable energy development – A discussion. *Renewable and Sustainable Energy Reviews*, 69(June 2015), 1170–1181. <https://doi.org/10.1016/j.rser.2016.09.137> Elsevier.
- Shearer, C., Mathew-Shah, N., & Myllyvirta, L. (2018). *Boom and bust 2018*. <https://doi.org/10.1016/j.jacr.2018.08.035>.
- Shearer, C., et al. (2020). *Boom and bust 2020: Tracking the global coal plant pipeline*. 28 Available at: https://endcoal.org/wp-content/uploads/2020/03/BoomAndBust_2020_English.pdf.
- Shen, J., et al. (2019). Does migration of pollution-intensive industries impact environmental efficiency? Evidence supporting “Pollution Haven Hypothesis”. *Journal of Environmental Management*, 242(November 2018), 142–152. <https://doi.org/10.1016/j.jenvman.2019.04.072> Elsevier.
- Shukla, A. K., Sudhakar, K., & Baredar, P. (2017). Renewable energy resources in South Asian countries: Challenges, policy and recommendations. *Resource-Efficient Technologies*, 3(3), 342–346. <https://doi.org/10.1016/j.reffit.2016.12.003> Elsevier B.V.
- Skoczkowski, T., et al. (2018). Impact assessment of climate policy on Poland's power sector. *Mitigation and Adaptation Strategies for Global Change*, 23(8), 1303–1349. <https://doi.org/10.1007/s11027-018-9786-z>.
- Steckel, J. C., Edenhofer, O., & Jakob, M. (2015). Drivers for the renaissance of coal. *Proceedings of the National Academy of Sciences of the United States of America*, 112(29), E3775–E3781. <https://doi.org/10.1073/pnas.1422722112>.
- Steffen, B., & Schmidt, T. S. (2019). A quantitative analysis of 10 multilateral development banks' investment in conventional and renewable power-generation technologies from 2006 to 2015. *Nature Energy*, 4(1), 75–82. <https://doi.org/10.1038/s41560-018-0280-3> Springer US.
- Sutton, M. (2015). Mongolia's 'third neighbor'. Available at: <https://www.japantimes.co.jp/opinion/2015/03/15/commentary/world-commentary/mongolias-third-neighbor/>.
- Svobodova, K., et al. (2020). Complexities and contradictions in the global energy transition: A re-evaluation of country-level factors and dependencies. *Applied Energy*, 265(February), 114778. <https://doi.org/10.1016/j.apenergy.2020.114778> Elsevier.
- The Ministry of Foreign Affairs of Japan (2017). Joint statement on deepening the Japan-Viet Nam extensive strategic partnership. Available at: <https://www.mofa.go.jp/files/000262573.pdf>.
- Thomas, V. F. (2021). *Indonesia's carbon tax plan: Will it fix the budget, the climate or neither?* The Jakarta Post Available at: <https://www.thejakartapost.com/news/2021/06/16/indonesias-carbon-tax-plan-will-it-fix-the-budget-the-climate-or-neither.html>.
- Tollefson, J. (2018). Carbon's future in black and white: There are reasons to be optimistic that the world will break its addiction to fossil fuels. But time is running out. *Nature*, 422–425.
- Trencher, G., Downie, C., et al. (2020a). Divestment trends in Japan's international coal businesses. *Renewable and Sustainable Energy Reviews*, 124(May 2019), 109779. <https://doi.org/10.1016/j.rser.2020.109779> Elsevier Ltd.
- Trencher, G., Rinscheid, A., et al. (2020b). Revisiting carbon lock-in in energy systems: Explaining the perpetuation of coal power in Japan. *Energy Research and Social Science*, 69(April), 101770. <https://doi.org/10.1016/j.erss.2020.101770> Elsevier.
- Tritto, A. (2021). China's Belt and Road Initiative: From perceptions to realities in Indonesia's coal power sector. *Energy Strategy Reviews*, 34. <https://doi.org/10.1016/j.esr.2021.100624> Elsevier Ltd.
- Uddin, M., Zakkour, D. P., & Lewington, P. (2019). Update on international finance for coal-fired power plants. Available at: <https://www.iea-coal.org/report/update-on-international-finance-for-coal-fired-power-plants-muhammad-uddin-dr-paul-zakkour-and-paul-lewington/>.
- Ulagan, Y. (2021). Belt and road initiative: Opportunities and challenges for Mongolia. *Asia-Pacific Journal: Japan Focus*, 19(3), 1–12.
- Umbach, F., & Yu, K. (2010). China's expanding overseas coal power industry. Available at: <https://eucers.com/wp-content/uploads/2019/03/strategy-paper-11.pdf>.
- UNDP (2018). opportunities for carbon pricing in Viet Nam. Available at: https://www.vn.undp.org/content/vietnam/en/home/library/environment_climate/opportunities-for-carbon-pricing-in-viet-nam.html.
- UNFCCC (2015). *Nationally Determined Contributions (NDCs)*. UNFCCC Available at: <https://unfccc.int/process-and-meetings/the-paris-agreement/the-paris-agreement/nationally-determined-contributions-ndcs>.
- UNFCCC (2019). *Study on the introduction of carbon pricing instruments in Pakistan*. (July), 1–147 Available at: <https://unfccc.int/sites/default/files/resource/Pakistanstudy23.05.2019-finalizeddraft.pdf>.
- UNFCCC (2020). All NDCs, NDC Registry (interim). Available at: <https://www4.unfccc.int/sites/NDCStaging/Pages/All.aspx> (Accessed: 6 July 2020).
- Wall, R., et al. (2019). Which policy instruments attract foreign direct investments in renewable energy? *Climate Policy*, 19(1), 59–72. <https://doi.org/10.1080/14693062.2018.1467826> Taylor & Francis.
- Whitaker, R. G., et al. (2020). Coincidence analysis: A new method for causal inference in implementation science. *Implementation Science*, 15(1), 1–10. <https://doi.org/10.1186/s13012-020-01070-3>.
- Woods, L. E. (2017). *As construction begins on Java's Batang coal plant, a divided community faces environmental problems*. Mongabay Available at: <https://news.mongabay.com/2017/01/as-construction-begins-on-javas-batang-coal-plant-a-divided-community-faces-environmental-problems/>.
- World Bank (2019). *GNI per capita*. The World Bank Available at: <https://datahelpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups>.
- World Bank Group (2017). *Regulatory indicators for sustainable energy*. World Bank Group, 239 Available at: <https://rise.worldbank.org/scores>.
- World Bank Group (2019). *Electricity consumption per year*. World Bank Group Available at: databank.worldbank.org/indicator/NY.GDP.PCAP.CD/1FF4a498/popular-Indicators.
- Yang, J., & Urpelainen, J. (2019). Is coal-fired power generation associated with rural electrification? A global analysis. *Energy Research and Social Science*, 58(September), 101274. <https://doi.org/10.1016/j.erss.2019.101274> Elsevier.
- Yao, L., Andrews-Speed, P., & Zia, M. (2018). *China-Indonesia coal relationship: A new phase under the Belt & Road Initiative*. (ESI Policy Brief).
- Yildizhan, H. (2017). Incentive policies for coal plants in Turkey. *Thermal Science*, 21(5), 1917–1924. <https://doi.org/10.2298/TSC1160428004Y>.
- Yuan, J., et al. (2018). Electric power investment risk assessment for belt and road initiative nations. *Sustainability (Switzerland)*, 10(9), 1–21. <https://doi.org/10.3390/su10093119>.
- Zafar, U., et al. (2018). An overview of implemented renewable energy policy of Pakistan. *Renewable and Sustainable Energy Reviews*, 82(September 2017), 654–665. <https://doi.org/10.1016/j.rser.2017.09.034> Elsevier Ltd.
- Zeng, S., et al. (2017). A review of renewable energy investment in the BRICS countries: History, models, problems and solutions. *Renewable and Sustainable Energy Reviews*, 74(March), 860–872. <https://doi.org/10.1016/j.rser.2017.03.016> Elsevier Ltd.
- Zhang, M. M., et al. (2019). Evaluating uncertain investment decisions in low-carbon transition toward renewable energy. *Applied Energy*, 240(September 2018), 1049–1060. <https://doi.org/10.1016/j.apenergy.2019.01.205> Elsevier.
- Zhao, S., & Alexandroff, A. (2019). Current and future struggles to eliminate coal. *Energy Policy*, 129(February), 511–520. <https://doi.org/10.1016/j.enpol.2019.02.031>.